



CHEMISTRY

Paper 1 Multiple Choice

9729/01

17 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

Multiple Choice Answer Sheet

Write your name, PDG and NRIC / FIN number, **including** the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 PRELIM

Exam Details: H2 Chemistry / Paper 1

Date: 17/09/2018

This document consists of **13** printed pages.

1 Use of the Data Booklet is relevant to this question.

P and *Q* are atoms of different elements.

P has nine fewer protons than *Q*.

P has one more unpaired electron than *Q*.

What are the identities of *P* and *Q*?

- 1 *P* is carbon and *Q* is phosphorus.
- 2 *P* is nitrogen and *Q* is sulfur.
- 3 *P* is oxygen and *Q* is chlorine.

- A** 1 only
B 2 only
C 1 and 3 only
D 2 and 3 only

2 How many subshells and orbitals are there in the third shell?

	subshells	orbitals
A	2	4
B	2	8
C	3	5
D	3	9

3 An ionic compound *E*

- has an empirical formula NH_2O ,
- reacts with NaOH(aq) to produce ammonia gas

How many electrons are present in the anion of *E*?

- A** 23 **B** 24 **C** 31 **D** 32

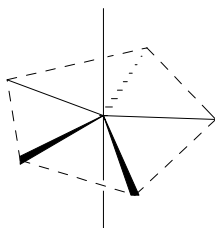
- 4 The shape of the PCl_5 molecule in the gas phase is trigonal bipyramidal.

The chlorine atom can be replaced one at a time by fluorine atoms, the axial chlorine atoms being replaced first, then the equatorial atoms.

Which molecule does **not** have a dipole moment?

- A PCl_4F B PCl_3F_2 C PCl_2F_3 D PClF_4

- 5 IF_7 is an interhalogen compound where the central atom is surrounded by seven pairs of electrons arranged in an unusual pentagonal bipyramidal shape. A pentagonal bipyramidal arrangement is shown below.



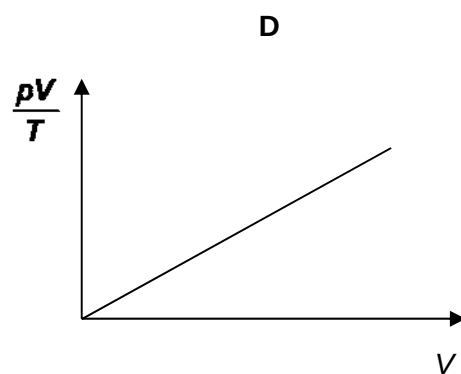
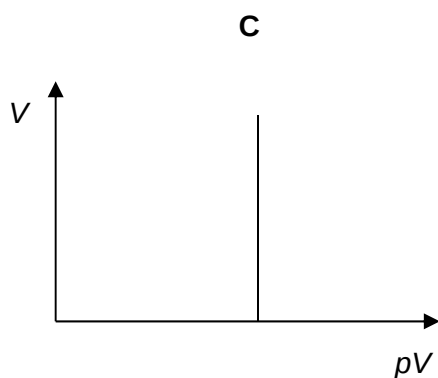
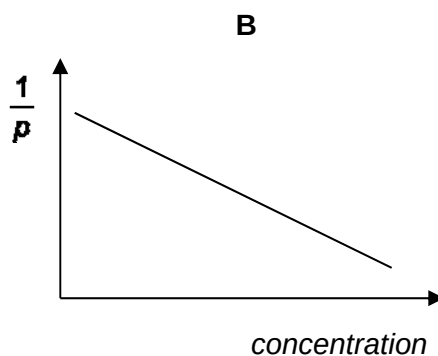
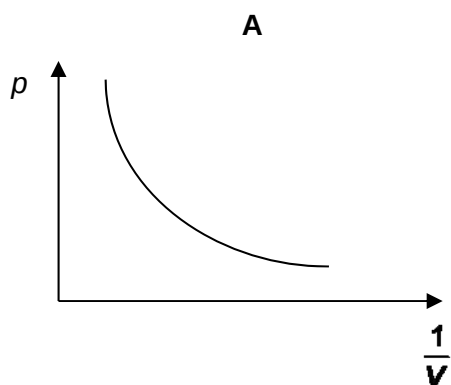
Which molecule has the same maximum number of atoms in a plane as a molecule of IF_7 ?

- A C_2H_4 B N_2H_4 C SiCl_4 D SF_6

- 6 Which statement about the molecule N_2F_2 is correct?

- A It contains dative bonds.
 B It contains four unpaired electrons.
 C It has cis–trans isomers.
 D It is linear in shape.

- 7 Which of the following diagrams correctly describes the behavior of a fixed mass of an ideal gas at constant T ?



- 8 The information relates to element Z.

- Z is in Period 3 of the Periodic Table.
- Z has a lower electrical conductivity than Mg.
- A Z atom has a half-filled subshell in its ground state.
- Z forms an acidic oxide on exposure to air.

What is Z?

A Na

B Si

C P

D Cl

- 9 A powder was known to be either a single sodium halide or a mixture of two sodium halides.

A sample of the powder was dissolved in water and aqueous silver nitrate added. A precipitate was formed, which, on addition of excess aqueous ammonia, partly dissolved leaving a yellow solid.

What did the powder consist of?

- A NaBr only
- B NaI only
- C a mixture of NaCl and NaBr
- D a mixture of NaCl and NaI

- 10 Antimony can be produced in a two-stage process from its ore stibnite, Sb_2S_3 .

The ore is first roasted in oxygen, producing Sb_4O_6 and SO_2 .

The Sb_4O_6 is then reduced by carbon, producing Sb and CO_2 .

What volume of CO_2 , measured at room temperature and pressure, is produced on processing 10 moles of Sb_2S_3 ?

- A 15 dm³ B 180 dm³ C 360 dm³ D 720 dm³

- 11 Most elements contain more than one isotope, atoms that have the same chemical properties but different masses.

Which expression correctly defines the term *relative atomic mass* of an element?

- A the mass of 1 atom of an element relative to the mass of 1 atom of ^{12}C
- B the mass of 1 mole of atoms of an element divided by 6.02×10^{23}
- C the mass of 1 mole of atoms of an element relative to the mass of 1 atom of ^{12}C
- D the mass of 1 mole of atoms of an element relative to the mass of 1 mole of ^{12}C atoms

- 12** The lattice energies of the compounds caesium chloride, caesium fluoride, sodium chloride and sodium fluoride are given below.

Which value corresponds to the lattice energy of caesium chloride?

- A** -661 kJ mol^{-1} **B** -747 kJ mol^{-1} **C** -780 kJ mol^{-1} **D** -918 kJ mol^{-1}

- 13** Clouds and rain form when air saturated with water vapour cools.



What are the correct signs of ΔH and ΔS for this reaction?

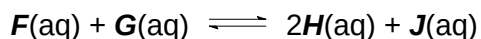
- A** both negative
B both positive
C ΔH negative, ΔS positive
D ΔH positive, ΔS negative

- 14** Which suggested mechanism is consistent with the experimentally obtained rate equation?

	rate equation	suggested mechanism
1	$\text{rate} = k_1 [\text{H}_2\text{O}_2] [\text{I}^-]$	$\begin{aligned} \text{H}_2\text{O}_2 + \text{I}^- &\xrightarrow{\text{slow}} \text{H}_2\text{O} + \text{OI}^- \\ \text{OI}^- + \text{H}^+ &\xrightarrow{\text{fast}} \text{HOI} \\ \text{HOI} + \text{H}^+ + \text{I}^- &\xrightarrow{\text{fast}} \text{I}_2 + \text{H}_2\text{O} \end{aligned}$
2	$\text{rate} = k_2 [\text{H}_2] [\text{I}_2]$	$\begin{aligned} \text{H}_2 &\xrightarrow{\text{slow}} 2\text{H} \\ 2\text{H} + \text{I}_2 &\xrightarrow{\text{fast}} 2\text{HI} \end{aligned}$
3	$\text{rate} = k_3 [\text{HBr}]^2 [\text{O}_2]^{1/2}$	$\begin{aligned} 2\text{HBr} + \text{O}_2 &\xrightarrow{\text{fast}} 2\text{HBrO} \\ \text{HBrO} + \text{HBr} &\xrightarrow{\text{slow}} \text{H}_2\text{O} + \text{Br}_2 \end{aligned}$
4	$\text{rate} = k_4 [\text{O}_3]^2 [\text{O}_2]^{-1}$	$\begin{aligned} \text{O}_3 &\xrightarrow{\text{fast}} \text{O}_2 + \text{O} \\ \text{O}_3 + \text{O} &\xrightarrow{\text{slow}} 2\text{O}_2 \end{aligned}$

- A** 1 only
B 1 and 4 only
C 2 and 3 only
D 1, 3 and 4 only

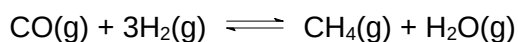
- 15 An equilibrium can be represented by the following equation.



In a certain 1 dm³ mixture, the equilibrium concentration of **G** is 10 mol dm⁻³.

What will be the new equilibrium concentration of **G** if 5 mol of pure **G** is dissolved in the mixture?

- A 15 mol dm⁻³
 B between 10 mol dm⁻³ and 15 mol dm⁻³
 C 10 mol dm⁻³
 D between 5 mol dm⁻³ and 10 mol dm⁻³
- 16 Which one of the following statements is correct about a reaction for which the equilibrium constant is independent of temperature?
- A The enthalpy change is zero.
 B Its rate constants do not vary with temperature.
 C There are equal number of moles of reactants and products.
 D The forward and reverse reactions are of the same order.
- 17 A mixture of CO and H₂ in a 1 : 3 molar ratio was introduced into a sealed vessel and heated to 1200 K.



At equilibrium it was found that the total pressure was 32 atm, and the mole fractions of CO and CH₄ were 0.19 and 0.12 respectively.

What is the value of the equilibrium constant, K_p ?

- A 4.0×10^{-4} B 1.0×10^{-3} C 1.3×10^{-1} D 4.0×10^{-1}

18 Values of two solubility products are given.

$$K_{\text{sp}} (\text{ZnF}_2) = 3.2 \times 10^{-2} \text{ mol}^3 \text{ dm}^{-9}$$

$$K_{\text{sp}} (\text{BaF}_2) = 1.6 \times 10^{-7} \text{ mol}^3 \text{ dm}^{-9}$$

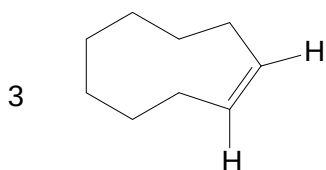
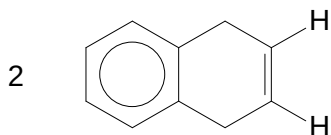
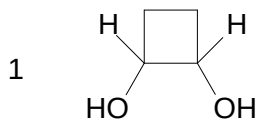
Solid zinc fluoride is shaken with water. The remaining solid is filtered off, leaving a saturated solution *M*.

Drops of dilute aqueous barium nitrate are added to *M* until barium fluoride just precipitates.

Which row of the table is correct?

	$[\text{F}^-]$ in <i>M</i> / mol dm^{-3}	$[\text{Ba}^{2+}(\text{aq})]$ when BaF_2 just precipitates
A	2×10^{-1}	4×10^{-7}
B	2×10^{-1}	8×10^{-7}
C	4×10^{-1}	1×10^{-6}
D	4×10^{-1}	4×10^{-6}

19 Which of the following compounds exhibit cis–trans isomerism?



- A** 1 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3

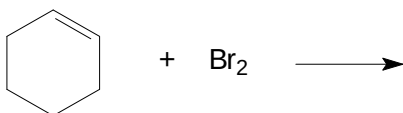
- 20** Ethane reacts with chlorine gas in the presence of ultraviolet light to form a mixture of products.

Which statements regarding this reaction are correct?

- 1 Both initiation and propagation steps involve homolytic fission.
- 2 Both initiation and propagation steps produce chlorine atom.
- 3 Both propagation and termination steps involve C–Cl bond formation.

- A** 1 only
B 2 only
C 2 and 3 only
D 1, 2 and 3 only

- 21** When cyclohexene is added to bromine in an inert organic solvent, there is a fast reaction that decolourises the bromine.

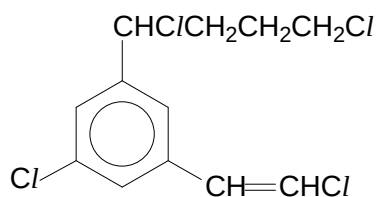


Which of the following statements is correct?

- 1 There is σ bond in cyclohexene that is formed by $1s-2sp^2$ orbital overlap.
- 2 There is a carbon atom that changed from being sp^2 hybridised in the reactant to being sp^3 hybridised in the intermediate formed during the reaction.
- 3 There are a total of 4 stereoisomers in the product mixture.

- A** 1 only
B 1 and 2 only
C 2 and 3 only
D 1, 2 and 3

22 Compound **K** has the structure as shown below.

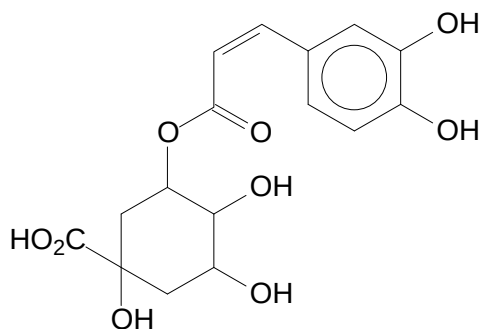


1 mol of compound **K** was heated with an excess of aqueous sodium hydroxide. The resulting reaction mixture was then cooled and acidified with dilute nitric acid. Excess aqueous silver nitrate is then added to it.

How many moles of silver chloride will be precipitated out?

- A** 0 **B** 1 **C** 2 **D** 4

23 Chlorogenic acid occurs naturally in coffee.

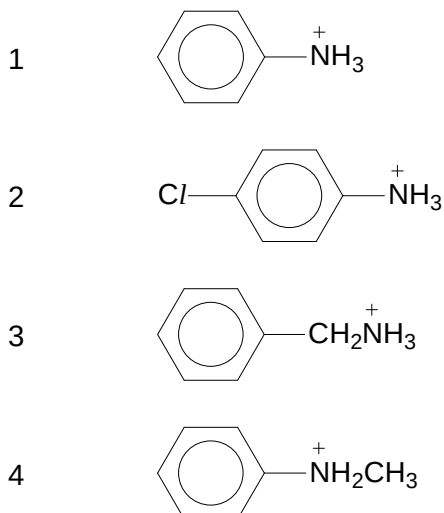


chlorogenic acid

Which of the following observations is correct?

- A** It reacts with 2,4–dinitrophenylhydrazine to give an orange precipitate.
B It reacts with hot, acidified potassium manganate(VII) to produce a gas which gives a white precipitate with limewater.
C It reacts with aqueous bromine to incorporate up to 7 atoms of bromine in each molecule.
D It reacts with excess sodium metal to produce 6 mol of hydrogen gas for each mole of the molecule.

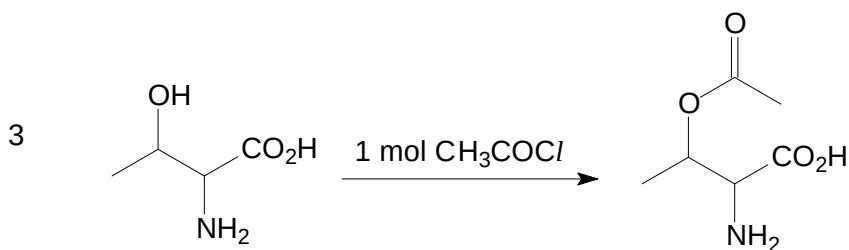
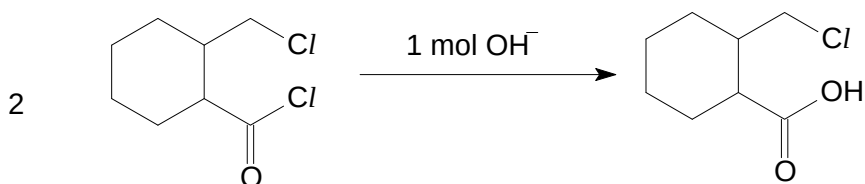
24 Consider the following four ions.



Which of the following shows the given ions arranged in order of **decreasing** pK_a ?

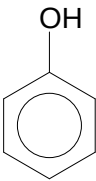
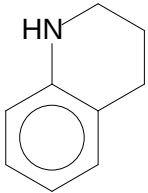
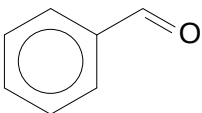
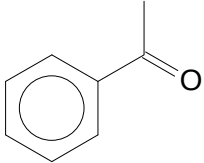
- | | | | | |
|----------|---|---|---|---|
| A | 3 | 4 | 1 | 2 |
| B | 3 | 4 | 2 | 1 |
| C | 4 | 3 | 1 | 2 |
| D | 4 | 2 | 3 | 1 |

25 Which of the reactions would **not** give the product shown?

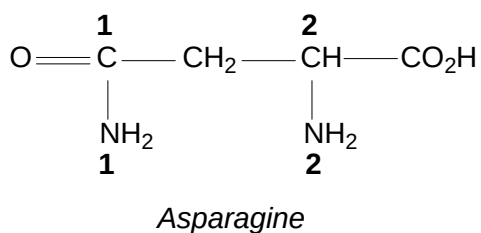


- A** 2 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 1, 2 and 3

- 26 Which of the following reagents and conditions can be used to distinguish the pair of compounds in each set?

	<u>Compound 1</u>	<u>Compound 2</u>	<u>Reagent and condition</u>
A			$\text{Br}_2(\text{aq})$, room temperature
B	$\text{CH}_3\text{CH}_2\text{COCl}_3$	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$	$\text{I}_2(\text{aq})$, $\text{NaOH}(\text{aq})$, warm
C			$[\text{Ag}(\text{NH}_3)_2]^+$ in alkaline solution, warm
D	$\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3$	$\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$	$\text{KMnO}_4(\text{aq})$, dilute H_2SO_4 , heat

- 27 *Asparagine* can be synthesised from the central metabolic pathway intermediates in human and is not required in the diet.

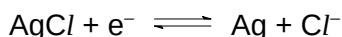


Which statements are correct?

- 1 *Asparagine* contains two sp^2 hybridised carbon atoms.
 - 2 The single C1–N1 bond is shorter than the single C2–N2 bond.
 - 3 One mole of *asparagine* requires two moles of HCl for complete neutralisation.
- A 1 only
 B 1 and 2 only
 C 1 and 3 only
 D 2 and 3 only

28 Use of the Data Booklet is relevant to this question.

The $\text{AgCl} | \text{Ag}$ reference electrode is based on the following reaction:



The reduction potential of the $\text{Fe}^{2+} | \text{Fe}$ couple was found to be -0.64 V when measured using the $\text{AgCl} | \text{Ag}$ reference electrode.

What is the reduction potential of the $\text{AgCl} | \text{Ag}$ electrode with respect to the standard hydrogen electrode?

- A -0.20 V B -1.10 V C $+0.20 \text{ V}$ D $+1.10 \text{ V}$

29 When a large current was passed through acidified aqueous copper(II) sulfate, there was simultaneous liberation, at the cathode, of $x \text{ mol}$ of copper and $y \text{ dm}^3$ of hydrogen gas, measured at standard temperature and pressure.

How many moles of electrons passed through the solution?

- A $x + \frac{y}{22.7}$ B $x + \frac{2y}{22.7}$ C $x + \frac{4y}{22.7}$ D $2x + \frac{2y}{22.7}$

30 Use of the Data Booklet is relevant to this question.

Lithium cells are used in digital watches, hearing aids and mobile phones.

They consist of lithium and manganese(IV) oxide in non-aqueous pastes enclosed between sheets of stainless steel.

The electrode reactions are as follows.

negative electrode: $\text{Li(s)} \longrightarrow \text{Li}^+(\text{s}) + \text{e}^-$

positive electrode: $\text{Li}^+(\text{s}) + \text{MnO}_2(\text{s}) + \text{e}^- \longrightarrow \text{LiMnO}_2(\text{s})$

The capacity of a cell is given in amp hours (Ah), which is the amount of constant current that can be drawn from the cell in one hour before it becomes discharged.

Which expression will calculate the minimum mass of lithium required to make a cell with a capacity of 1.8 Ah ?

- A $\frac{1.8 \times 3600 \times 6.9}{96500} \text{ g}$
- B $\frac{1.8 \times 3600 \times 6.9}{1.6 \times 10^{-19}} \text{ g}$
- C $\frac{1.8 \times 96500 \times 6.9}{3600} \text{ g}$
- D $\frac{1.8 \times 1.6 \times 10^{-19}}{6.9 \times 3600} \text{ g}$

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	D	16	A
2	D	17	A
3	B	18	C
4	B	19	B
5	A	20	D
6	C	21	B
7	C	22	C
8	C	23	B
9	D	24	A
10	C	25	D
11	D	26	C
12	A	27	B
13	A	28	C
14	D	29	D
15	B	30	A

A	7
B	7
C	8
D	8